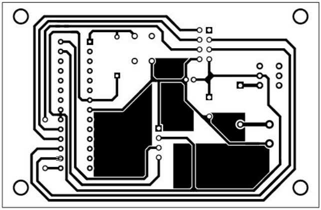

Fig. 8: Actual-size PCB layout of the receiver

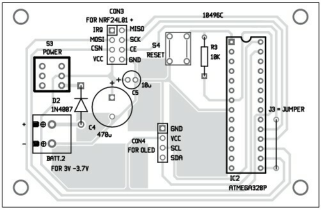

Fig. 9: Components layout for the PCB in Fig. 8

will last for 150/8.68571 = 17 days

Current consumption (receiver) is $\{(10\times32)+(32\times8.0)\}/42$ = 13.714mA seconds

$=(13.714\times60\times60\times24)/3600$

$=329.143\text{mAh}$

On a 3.7-volt, 150mAh button cell, it will last for about 150/329 = 27 minutes.

Display consumes more current.

Note that nRF24L01 radios operate off at maximum 3.6 volts. Sometimes, when the Li-ion battery is fully charged, the voltage may go as high as 4.01 volts, so the radio may not work. Put a diode in series with the battery connection so that the voltage drops by 0.7 volt and the radios start working again.

Construction and testing

An actual-size PCB layout for the transmitter circuit is shown in Fig. 6 and its components layout in Fig. 7. PCB layout for the receiver circuit is shown in Fig. 8 and its components layout in Fig. 9. After assembling the circuits on respective PCBs, enclose these in suitable cabinets and place them some distance apart.

Connect the battery to the transmitter unit and measure the current. The current withdrawn is 6-7 mA while transmitting for two seconds and 30μA for the remaining 40 seconds. For periodicity, we used the WDT of the MCU, which is not very precise. It provides five incremental steps of 250ms, 500ms, 2, 4 and 8 seconds. That means, it will transmit once every $5\times8+2=42$ seconds. The DHT22 sensor requires a minimum of 2.8 volts to operate. Therefore put a fairly big capacitor (100-470μF) across its Vcc and GND pins. Else, you will get temperature and relative humidity values as zero. The sensor takes two seconds between readings. EFY

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EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at *source.efymag.com*